

Methodological Integrity Assessment

EMA200 Trend-Following Strategy

Strategy Summary

The strategy operates on 4-hour candles for ETH and BNB. A 200-period Exponential Moving Average (EMA) is used as the sole trend filter.

$$Signal_t = \begin{cases} \text{Bullish}, & Price_t > EMA_t \\ \text{Bearish}, & Price_t \leq EMA_t \end{cases}$$

Portfolio allocation follows:

- Both assets bullish $\rightarrow$ 50% ETH + 50% BNB
- Only ETH bullish $\rightarrow$ 100% ETH
- Only BNB bullish $\rightarrow$ 100% BNB
- Neither bullish $\rightarrow$ 100% USDT

The objective is to remain invested only during persistent uptrends while preserving capital during bearish market regimes.

1. Look-Ahead Bias Analysis

Look-ahead bias occurs when future market information is inadvertently used to generate historical trading decisions.

The implemented strategy avoids this issue because every signal is generated exclusively from information available at the decision timestamp.

The EMA is calculated recursively as

$$EMA_t = \alpha P_t + (1 - \alpha) EMA_{t-1}$$

where

$$\alpha = \frac{2}{N + 1}, \quad N = 200.$$

Consequently,

$$EMA_t = f(P_1, P_2, \dots, P_t)$$

and the trading decision is

$$Decision_t = g(P_1, P_2, \dots, P_t).$$

Future observations such as

$$P_{t+1}, \quad P_{t+2}, \quad P_{t+3}$$

are never referenced during signal generation.

The execution sequence is therefore

$$Historical\ Data \rightarrow EMA\ Calculation \rightarrow Signal\ Generation \rightarrow Trade\ Execution.$$

Since every trade is based solely on information that would have been available at time t , no observable look-ahead bias exists within the strategy.

2. Survivorship Bias Analysis

Survivorship bias occurs when a backtest excludes assets that failed, disappeared, or underperformed, thereby overstating historical performance.

The present strategy does not dynamically select assets from a changing cryptocurrency universe.

Instead, the trading universe is fixed before the backtest begins:

$$\mathcal{U} = \{ETHUSDT, BNBUSDT\}.$$

The same two assets remain in the universe throughout the entire testing period.

$$Backtest\ Start \rightarrow \{ETH, BNB\} \rightarrow Backtest\ End$$

No asset is introduced or removed based on future performance.

Because the universe remains constant and is specified *a priori*, the strategy does not exhibit survivorship bias arising from asset selection.

3. Data Snooping Bias Analysis

Data snooping bias occurs when numerous parameter combinations are tested and only the best-performing result is ultimately reported.

Typical examples include repeatedly testing

$$EMA_{20}, EMA_{50}, EMA_{100}, EMA_{150}, EMA_{200}, EMA_{250}, EMA_{300}$$

and selecting whichever configuration produces the highest historical return.

The present implementation contains only a single core parameter:

$$N = 200.$$

The EMA200 filter was adopted directly as a standard long-horizon trend benchmark rather than being selected through extensive optimization.

No evidence exists of:

- Grid-search optimization
- Hyperparameter tuning
- Genetic optimization
- Walk-forward parameter mining
- Historical curve fitting

The effective model complexity is therefore

$$k = 1$$

where k denotes the number of primary tunable parameters.

Because overfitting risk generally increases with model complexity,

$$Risk_{Overfit} \propto k,$$

the simplicity of the strategy substantially reduces exposure to data snooping bias.

While no empirical strategy can claim absolute immunity without long-term out-of-sample validation, the current framework exhibits a low probability of parameter overfitting.

Conclusion

The EMA200 trend-following framework demonstrates strong methodological simplicity and transparency.

Criterion	Assessment
Look-Ahead Bias	Not Detected
Survivorship Bias	Not Detected
Data Snooping Bias	Low Risk
Parameter Complexity	Very Low
Interpretability	High
Reproducibility	High

The primary limitation of the strategy is not statistical bias but the relatively short historical testing horizon. Additional validation across multiple market regimes would further strengthen confidence in future performance.